Lewis Structures of Molecules Part 1 (simple molecules)

- Draw each atom along with it’s valence electrons
- Make one Bond between the central atom and each terminal atom by pairing up electrons
- Pair electrons to make double bonds if necessarily to complete octet for all atoms (maximum of four bonds per atom)

Lewis Structures of Molecules Part 1 (simple molecules) Example: methane, CH_4

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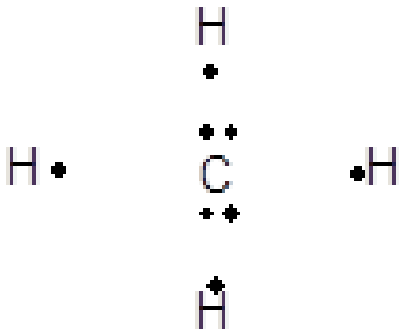
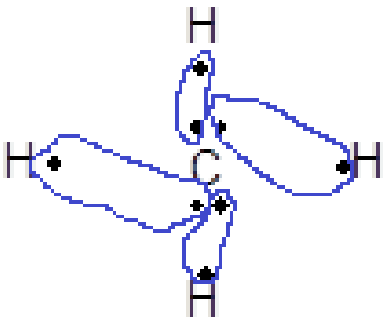
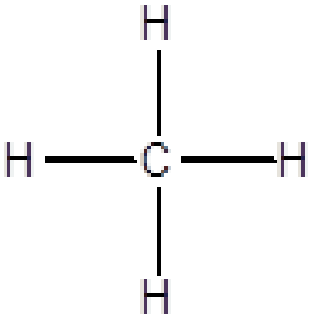


Lewis Structures of Molecules Part 1 (simple molecules) Example methane, CH_4

2.) Make one Bond between the central atom and each terminal atom by pairing up electrons





Lewis Structures of Molecules Part 1 (simple molecules)

3.) If necessarily pair electrons to make double (or triple) bonds to complete octet for all atoms to a maximum of four bonds per atom. (This is strictly true for elements of the 2nd period. We will cover exceptions later.)

Lewis structures of Molecules

Learn by doing!!!

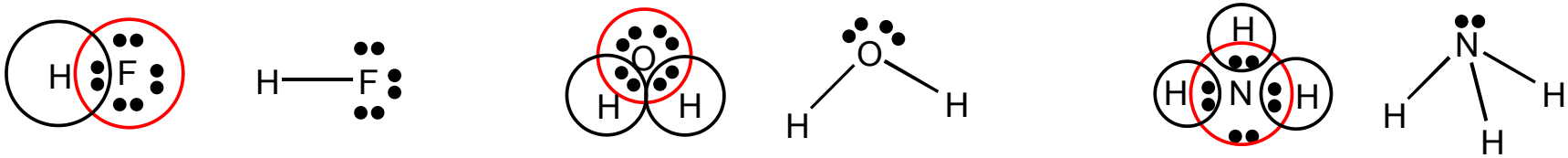
Write Lewis structures for the following molecules:

HF

H_2O

NH_3

Answers:



A bond is represented as a solid line between atoms.

Lewis structures and the central atom:

For the molecules CH_4, NH_3 and H_2O you might have wondered which atom would be the central atom. Note that hydrogen would not make a very good central atom since it would have to form more than one bond.

How about CO_2, HCN or N_2O though?

Lewis structures and the central atom:

1.) Usually, but not always, the least electron affine element is the central atom **<u>C</u>O_2, H<u>C</u>N or <u>N</u>$_2$O**

e.g. in oxo-anions such as $[CO_3]^{2-}$, $[ClO_4]^{-}$, $[PO_4]^{3-}$or $[SO_4]^{2-}$ the non oxygen atom is the central atom

Lewis structures and the central atom:

2.) Generally, if there are four atoms or more, small compact structures are preferred over linear arrangements **SO_3**

e.g. the molecule **SO_3** does not consist of a chain of four atoms but carries a central sulfur atom (less electronegative then oxygen) with three oxygen atoms surrounding it.

Lewis structures of Molecules

Learn by doing!!!

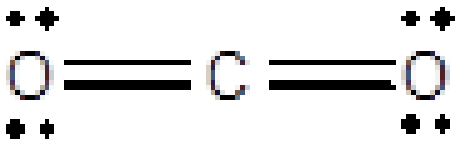
Write Lewis structures for the following molecules:

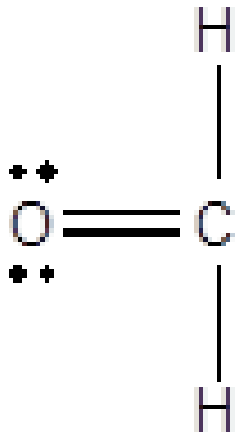
N_2

CO_2

CH_2O (formaldehyde)

HCN





Lewis structures of ions

The concept of Lewis structures also applies to polyatomic ions. We just have to take into account the extra electron(s) for anions, or the missing electron for cations.

$[NH_4]^+$

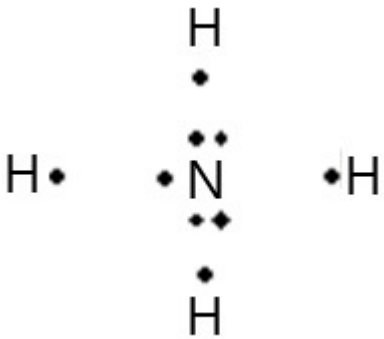
Lewis structures of ions

Write the Lewis structure for the ammonium ion!

$[NH_4]^+$

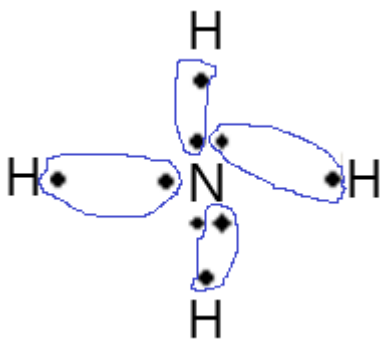
Lewis structures of ions

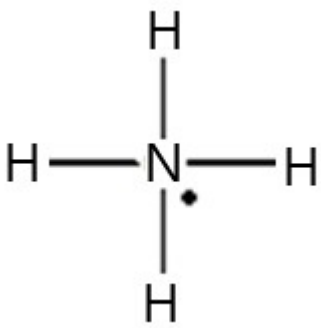
1.) Draw each atom along with it’s valence electrons



Lewis structures of ions

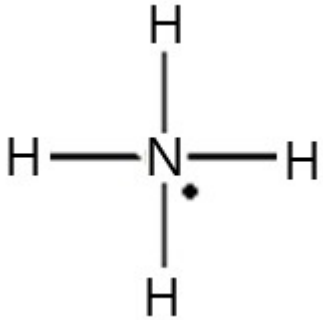
2.) Make one Bond between the central atom and each terminal atom by pairing up electrons

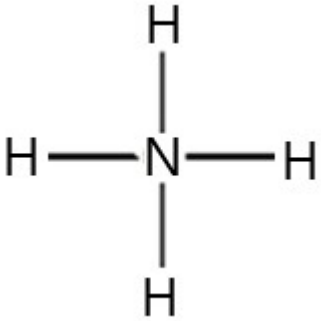




Lewis structures of ions

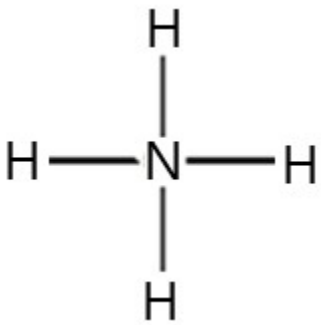
3.) If overall charge (n-) add n electrons to a terminal atom(s), <u>if overall charge n+ remove n electrons from the central atom</u>.

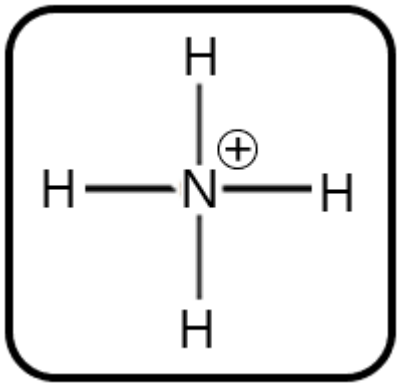




Lewis structures of ions

4.) Indicate the absence of that electron by placing a FORMAL CHARGE (circled positive charge sign) on the nitrogen atom.





Lewis structures of ions

Lets try an anion:
Draw the Lewis structure of the nitrite ion

Lewis structures of ions

Lets try an anion:
Draw the Lewis structure of the nitrite ion

$[NO_2]^-$

Lewis structures of ions

1.) Draw each atom along with it’s valence electrons

Lewis structures of ions

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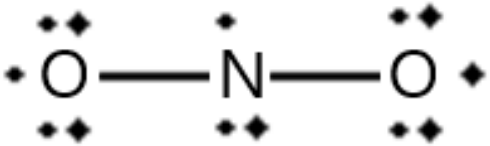
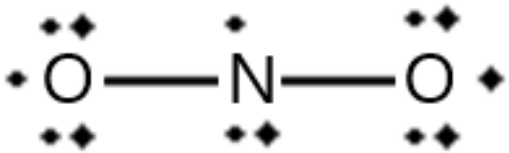
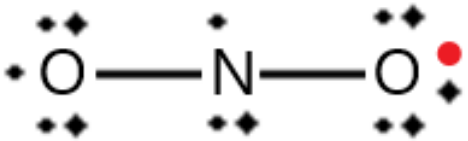


Lewis structures of ions

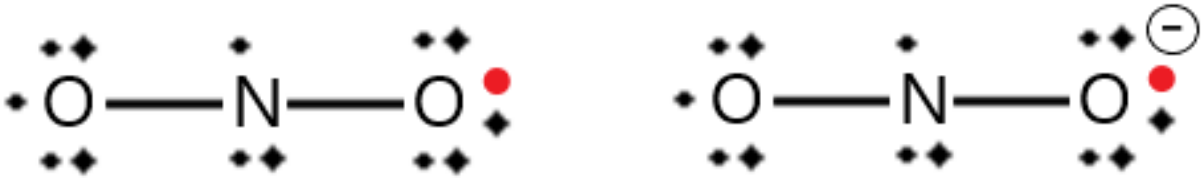
3.) <u>If overall charge (n-) add n electrons to a terminal atom(s)</u>, if overall charge n+ remove n electrons from the central atom.





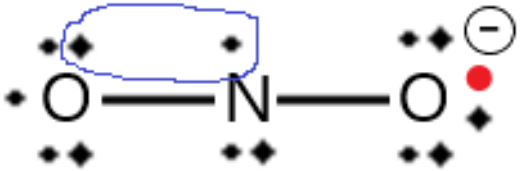
Lewis structures of ions

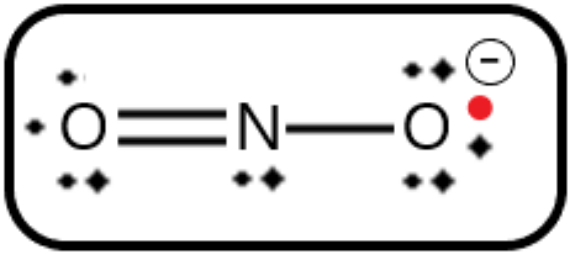
4.) Indicate the presence of the extra electron by placing a FORMAL CHARGE (circled negative charge sign) on the oxygen atom that received the extra electron.



Lewis structures of ions

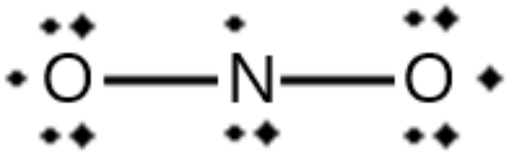
5.) If necessarily pair electrons to make double (or triple) bonds to complete octet for all atoms to a maximum of four bonds per atom.





Resonance

The nitrite example raises an important point. How did you decide which of the two oxygen atoms in the nitrite ion was to receive the extra electron



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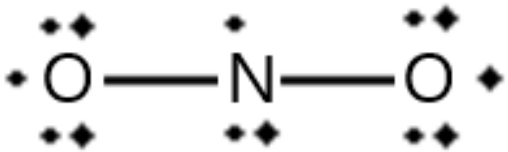
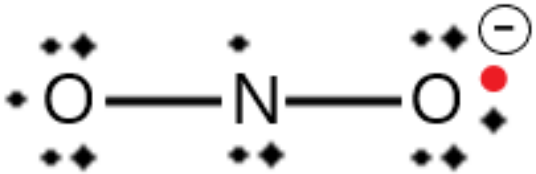
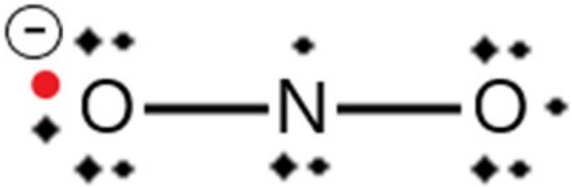
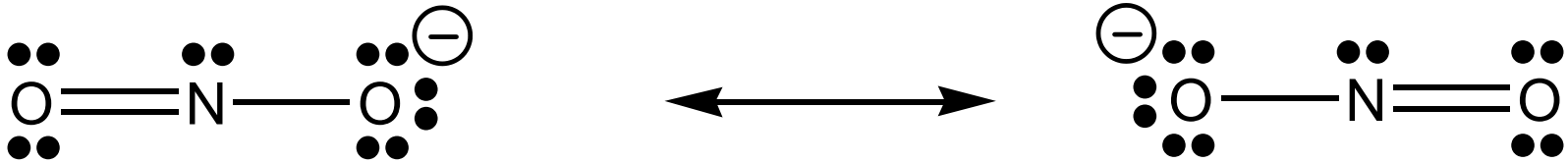






Resonance

The double bond would have ended up on the other side of the ion:



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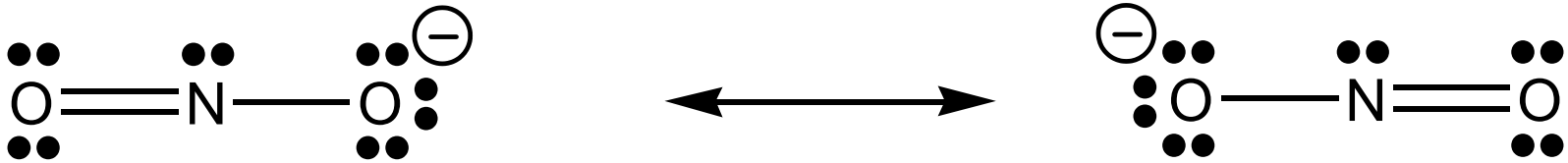
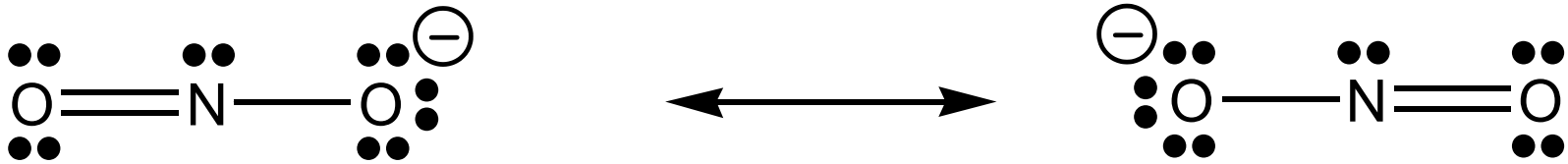


But what is the significance of this?

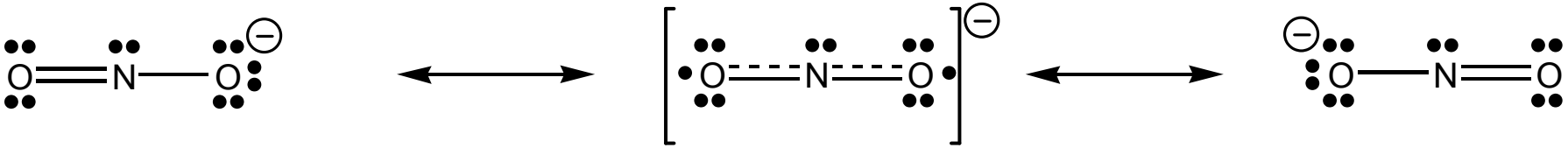
Resonance



Whenever we can draw more than one valid Lewis structure for a molecule, we have resonance and the two structures are referred to as resonance structures, Resonance is indicated by the double headed arrow.

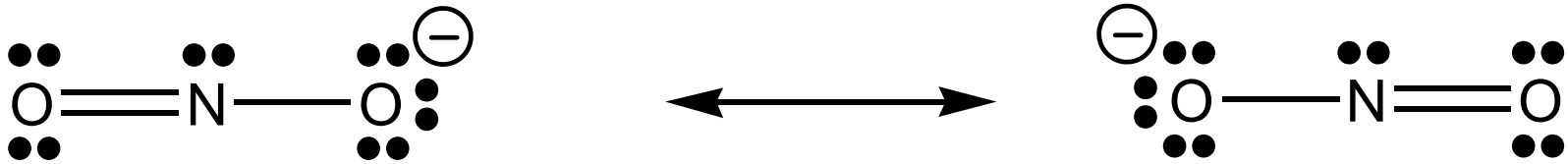
Resonance

The true structure, the one we actually observe, looks like something in between those two extremes and is called the ***resonance hybrid***.



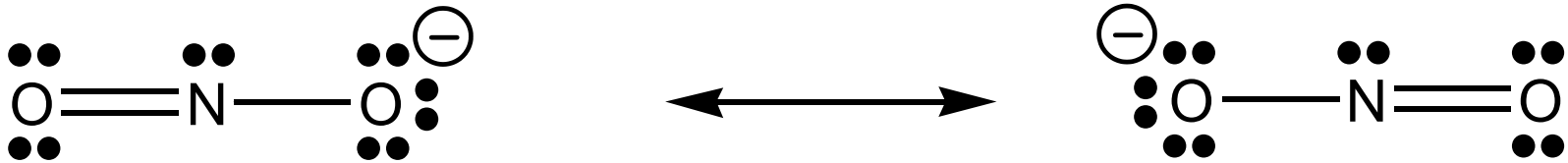
(The resonance hybris is not meant to be drawn as a valid Lewis structure)

Resonance



The two resonance structures are related by symmetry. They can both be superimposed upon another an are thus identical. This means they **contribute equally to the resonance hybrid** and are called **equivalent resonance structures.**

Resonance



Note that the electrons move orders of magnitudes faster than atoms, so if the ion was to flip over from one form to the other the electrons would have already flipped thousands of times between right and left form.

Atoms never move in resonance structures. Only electrons!

Resonance

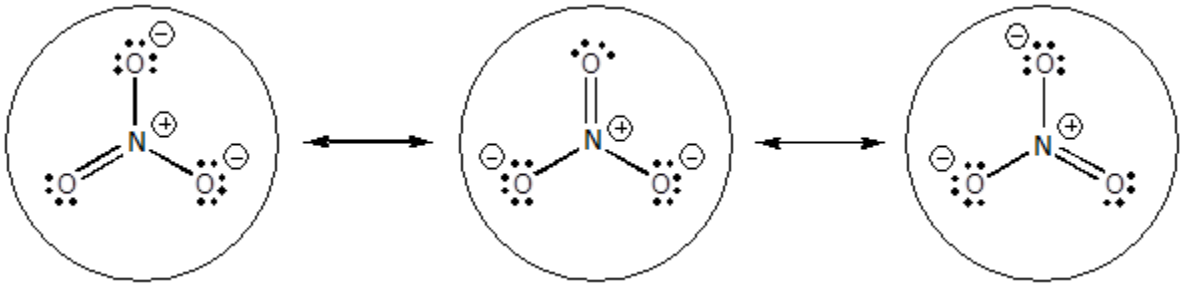
Two or more valid Lewis structures can be drawn for the same molecule or ion. The true structure is somewhere in between.

Consider:

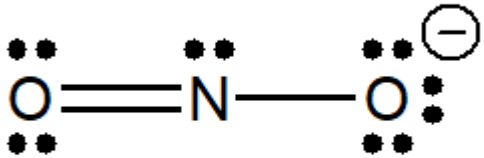
(i) NO_3^-

(ii) CO_3^{2-}

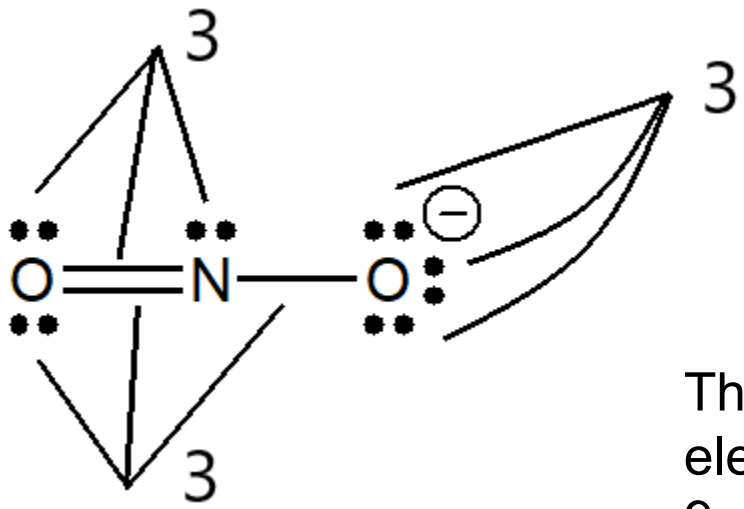
Resonance



Book keeping for valence electrons



It is essential that we draw valid Lewis structures. We can check if we have included all valence electrons (VE) by counting them and comparing that number to the # of VE our molecule or ion possesses: For nitrite:



1 N = 1 x 5 VE

2 O = 2 x 6 VE

-1 Charge = 1 VE

summ = 18 VE

There are a total of nine valence electron pairs in this Lewis structure: 9 x 2 = 18 VE

More bookkeeping: Calculating Formal Charges

Formal charge on an atom

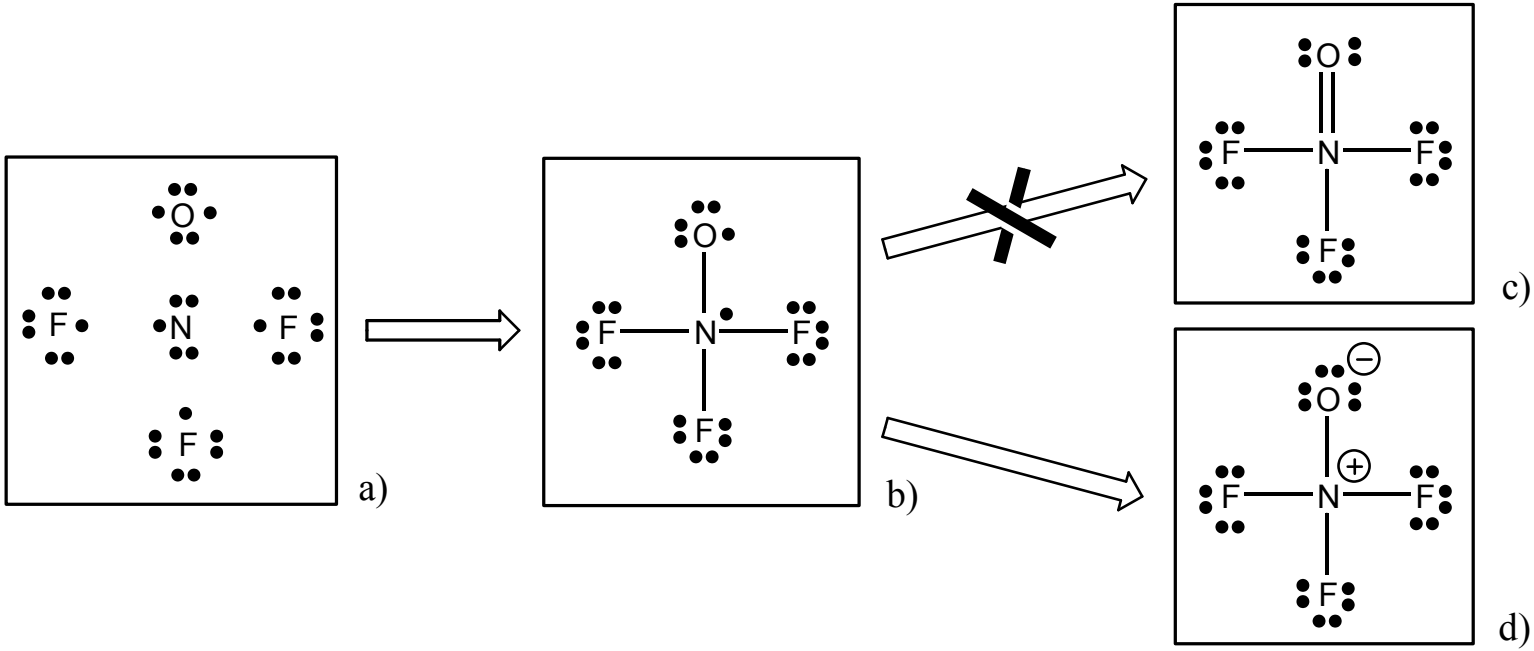
= # valence e^- – 0.5x(# bonding e^-) – (# lone pair e^-)

Consider,

NOF_3

DO NOT CONFUSE FC WITH OXIDATION STATE!

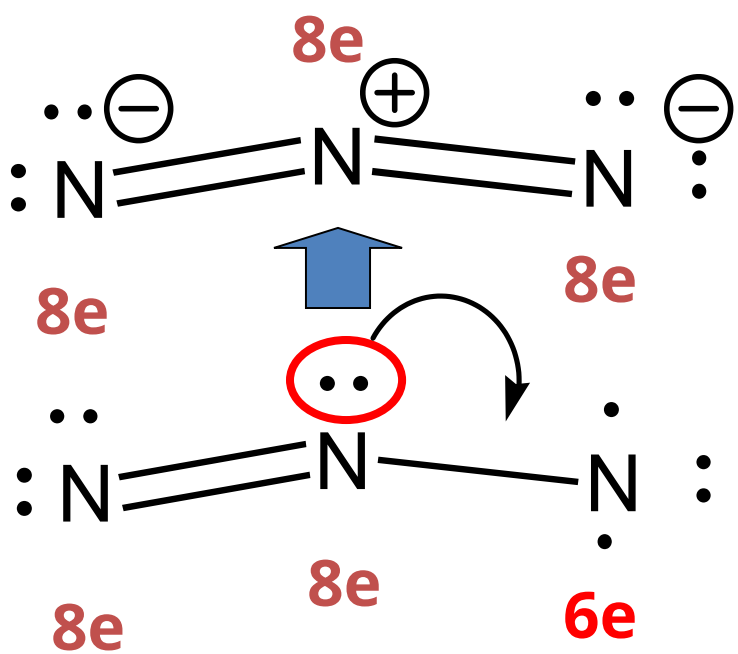
Formal Charges

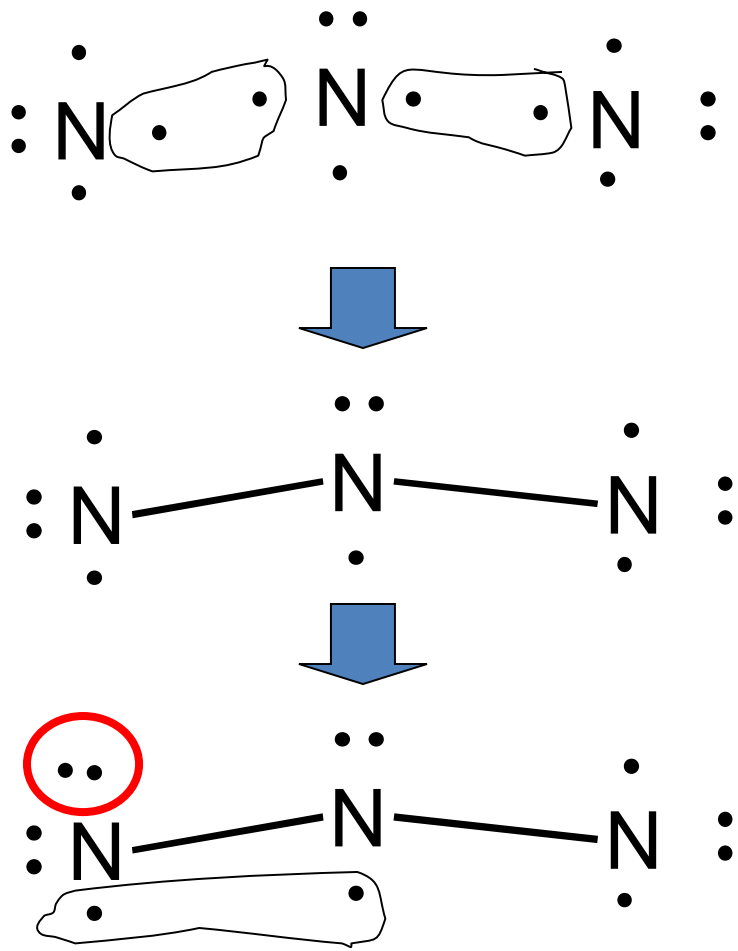


Draw the Lewis Structures of the azide ion

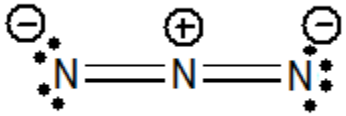
Azide (N_3^-)

3. If necessary, complete octets by forming additional bonds. This will result in formal charges on some atoms

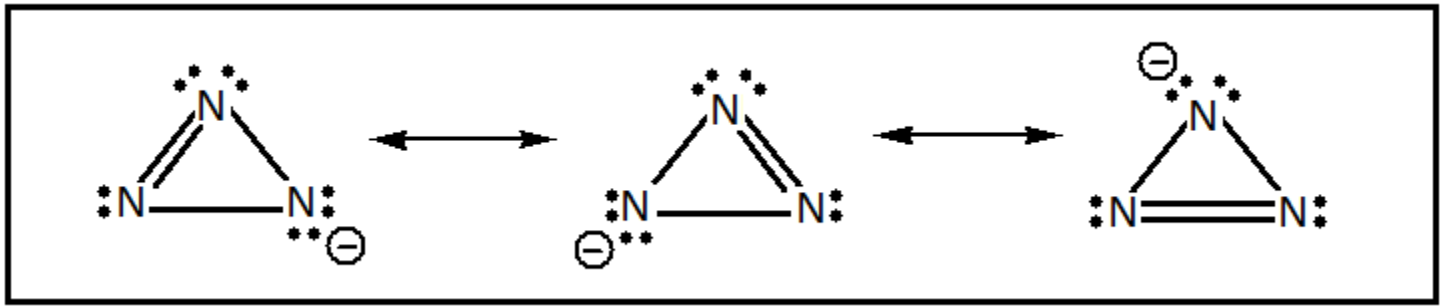




Resonance is not Isomerism!

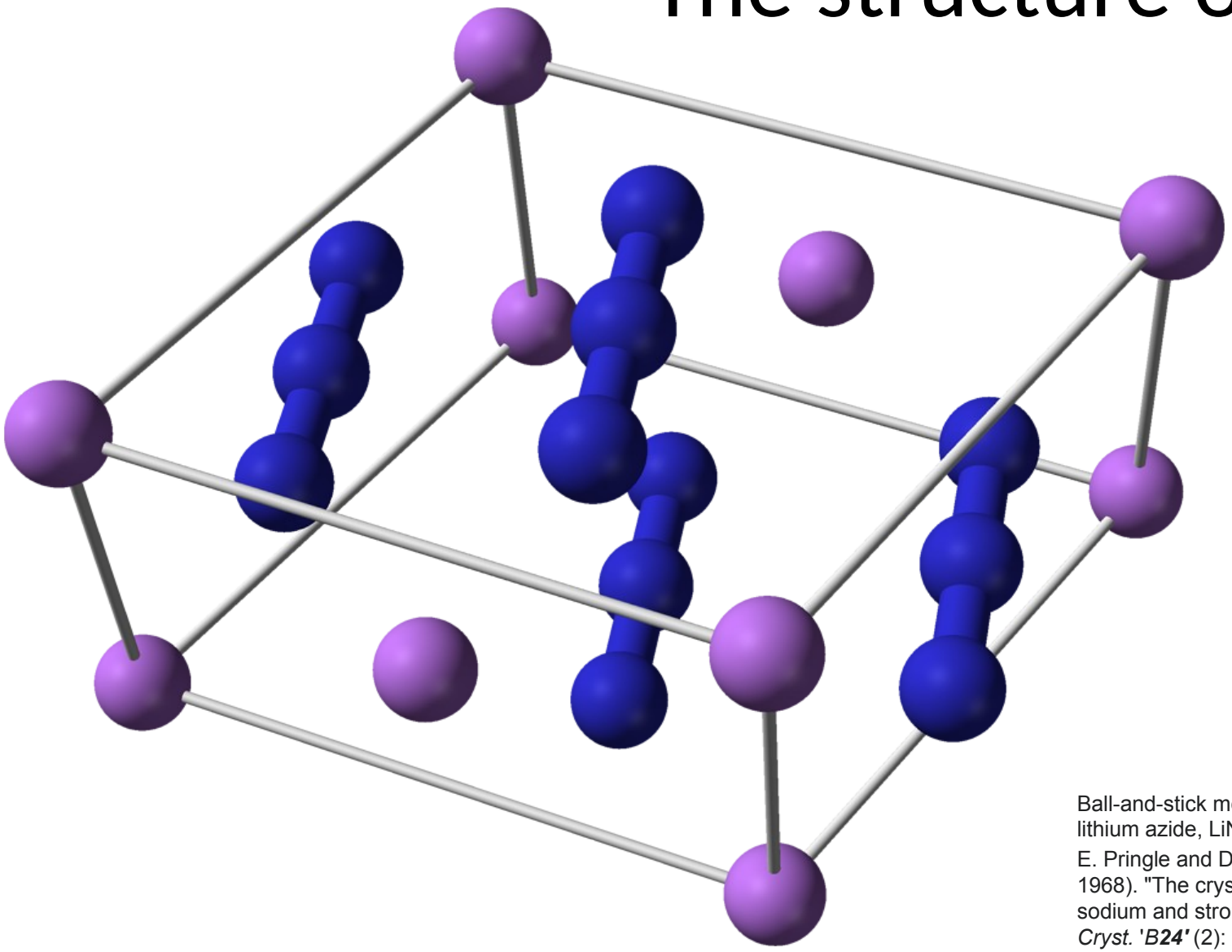


Isomers have the same number of atoms (same formula $[N_3]^-$) in a different arrangement (structural or geometrical)



Three equivalent resonance structures of a hypothetical cyclic isomer of $[N_3]^-$

The structure of $[N_3]^-$



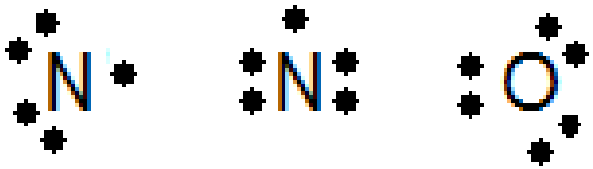
Ball-and-stick model of the unit cell of lithium azide, LiN_3. Structural data from G. E. Pringle and D. E. Noakes (February 1968). "The crystal structures of lithium, sodium and strontium azides". *Acta Cryst.* '***B24'*** (2): 262-269. DOI:10.1107/S0567740868000206 2.

Non-equivalent Resonance structures

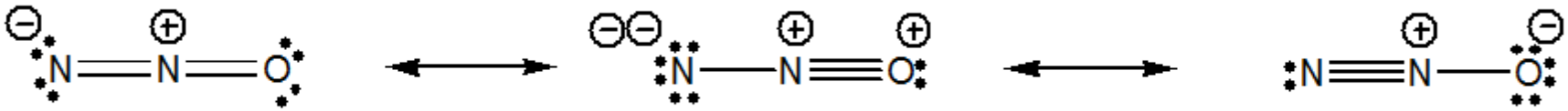
Consider: Dinitrogen monoxide

Draw all possible (valid) Lewis structures

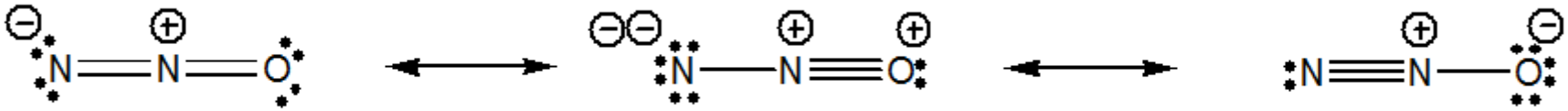
1.Draw each atom with its proper number of VE



Non-equivalent Resonance structures



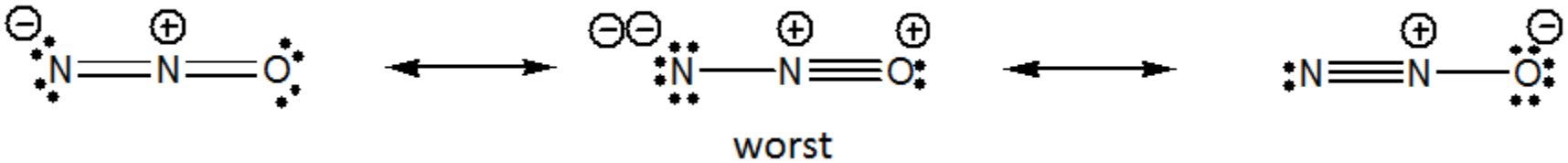
Non Equivalent Resonance Structures



Non–equivalent resonance structures do not contribute equally to the resonance hybrid.

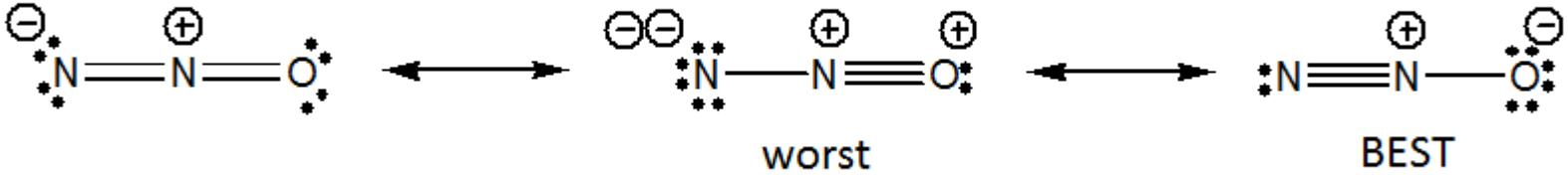
Finding the BEST Lewis structure among a set of non-equivalent resonance structures

1. The best Lewis structures has the lowest possible formal charges (e.g. zero formal charges!)



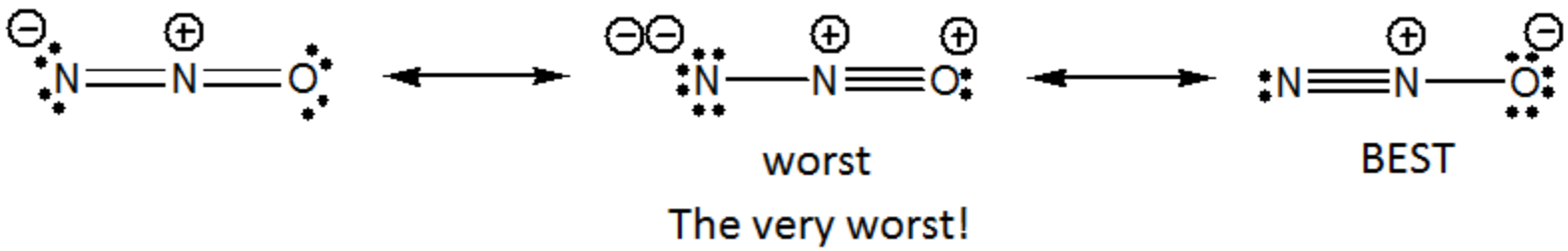
Finding the BEST Lewis structure among a set of non-equivalent resonance structures

2. If formal charges are necessarily the best Lewis structure will be the one that has the negative formal charge on the more electronegative atom and the positive charge on the less electronegative atom. (Electronegativity correlates with electron affinity.)



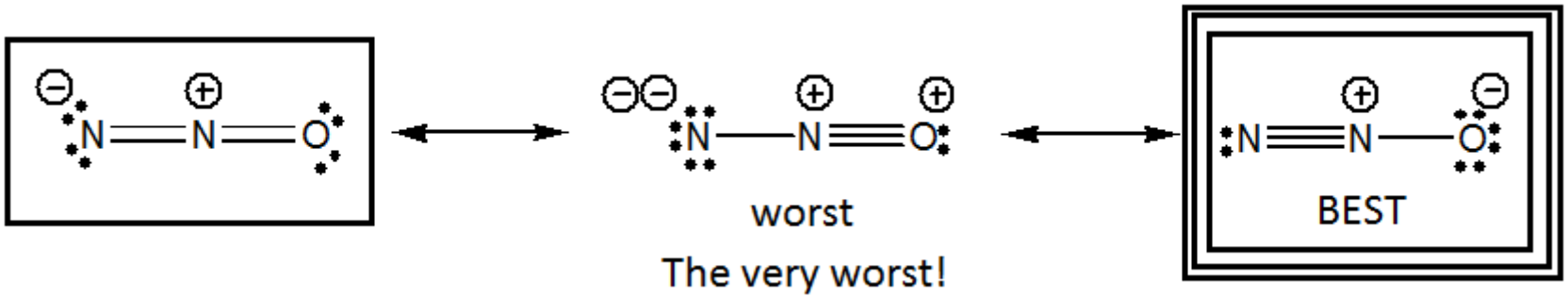
Finding the BEST Lewis structure among a set of non-equivalent resonance structures

3. Formal charges of the same sign on adjacent atoms are very unlikely.



What does this mean?

- The BEST resonance structure contributes most strongly to the resonance hybrid!



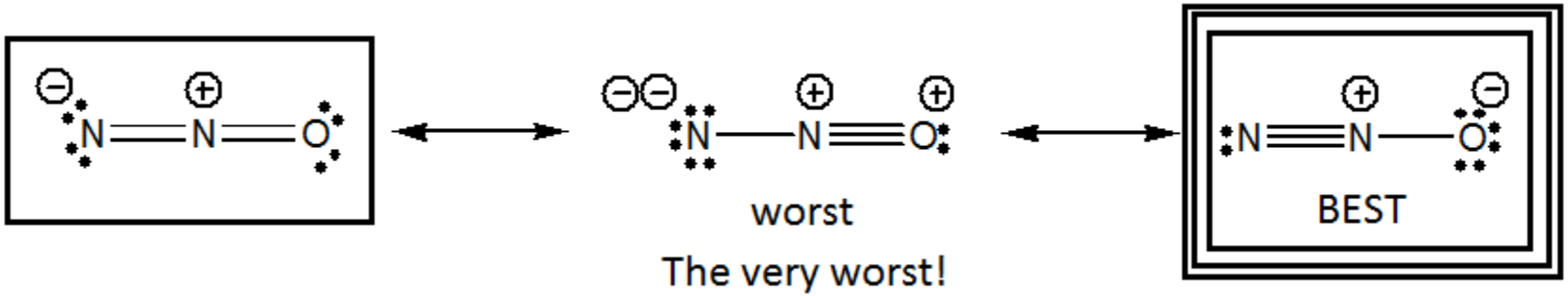
Relating structure to property:

Dinitrogen monoxide is an endothermic compound and is thermodynamically instable.

Propose the path by which it decomposes.

$$2N_2O \rightarrow 2N_2 + O_2$$

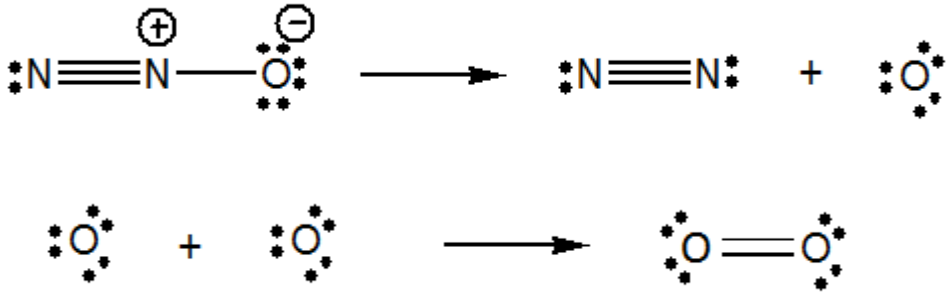
$$2N_2O \rightarrow 2NO + N_2$$



Relating structure to property:

Dinitrogen monoxide decomposed into the elements. Note that a chemical reaction is a rearrangement of atoms, some bonds need to break and new bonds form. The best resonance structure already contains two triple bonded N

$$2N_2O \rightarrow 2N_2 + O_2$$



Exceptions to the Octet Rule

- Odd-electron species (*eg* NO)
- Incomplete octets (*eg* BF_3)
- Expanded valence shells

  eg PCl_5, SF_6,

  SO_2, $(SO_4)^{2-}$ and $(ClO_4)^-$

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Most Important

Expanded Valence Shells

Row \ Group	13	14	15	16	17	18	
$n = 1$					H	He	maximum duet
$n = 2$	B	C	N	O	F	Ne	maximum octet
$n = 3$	Al	Si	P	S	Cl	Ar	Can exceeded an octet of electrons
$n = 4$	Ga	Ge	As	Se	Br	Kr	
$n = 5$	In	Sn	Sb	Te	I	Xe	
$n = 6$	Tl	Pb	Bi	Po	At	Rn	

Draw all possible Lewis structures for the sulfate ion:

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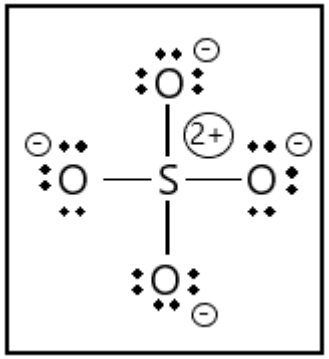
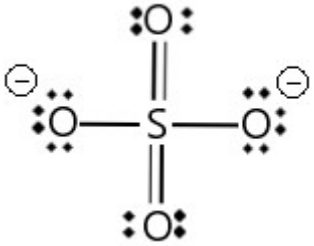
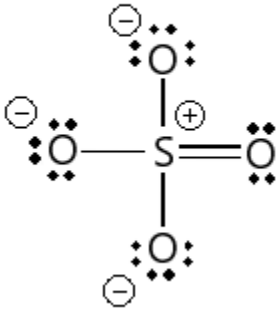
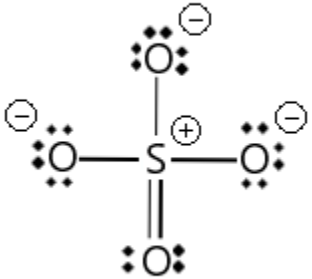


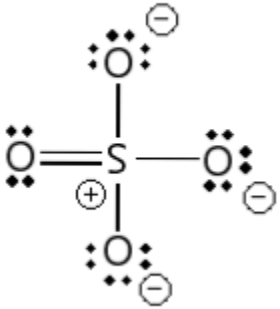


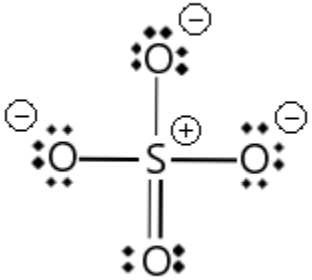
S obeys octet but high formal Charges!

Best structure (lowest formal charges 6 resonance structures possible)



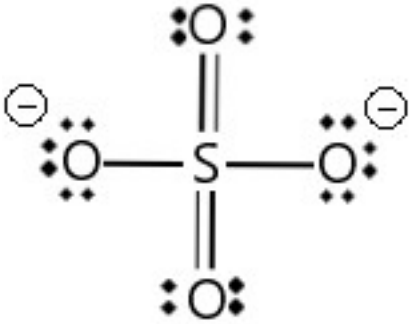






In CHEM 150

- For elements beyond the 2nd Period always minimize Formal Charges by forming multiple bonds.



How exceptions to the octet rule affect the chemistry of our atmosphere

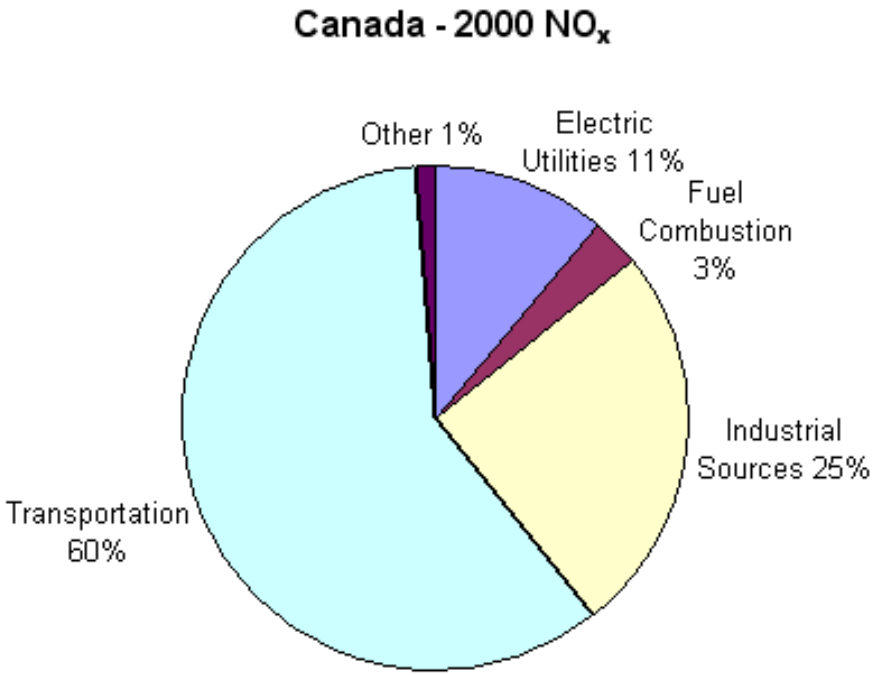
Composition of the Atmosphere

Gas	*Mixing ratio (mol/mol)*
Nitrogen (N_2)	0.78
Oxygen (O_2)	0.21
Nitrogen dioxide (NO_2)	1×10^{-7}
Hydroxyl radical (OH)	1×10^{-12}

- 78 % of the Earth's atmosphere consists of N_2
- Yet, N_2 does not play any role in the chemistry of the atmosphere?
- Explain using Lewis structure

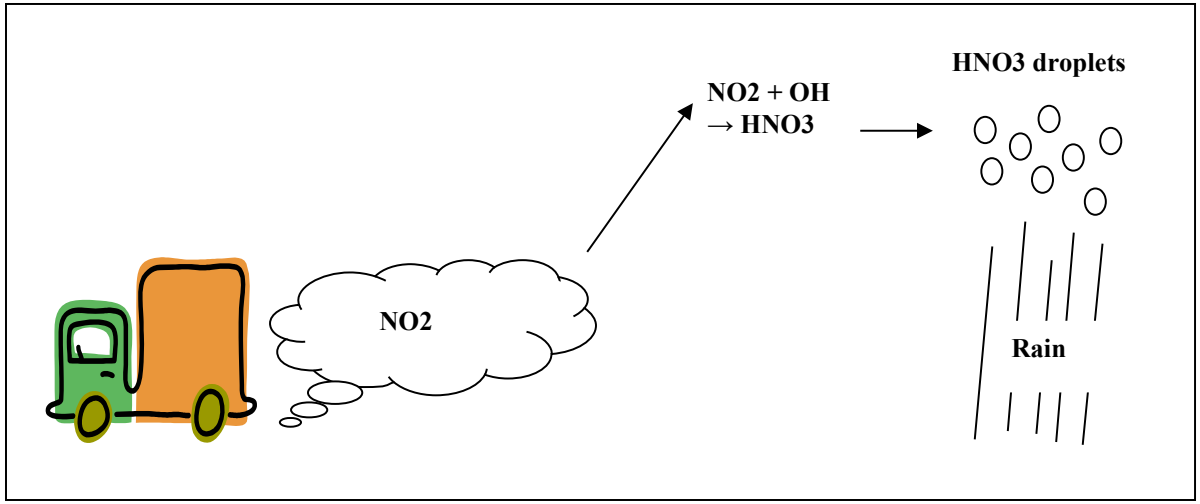
OH makes up a very small amount of atmosphere
Yet it is the most important molecule in terms of chemistry

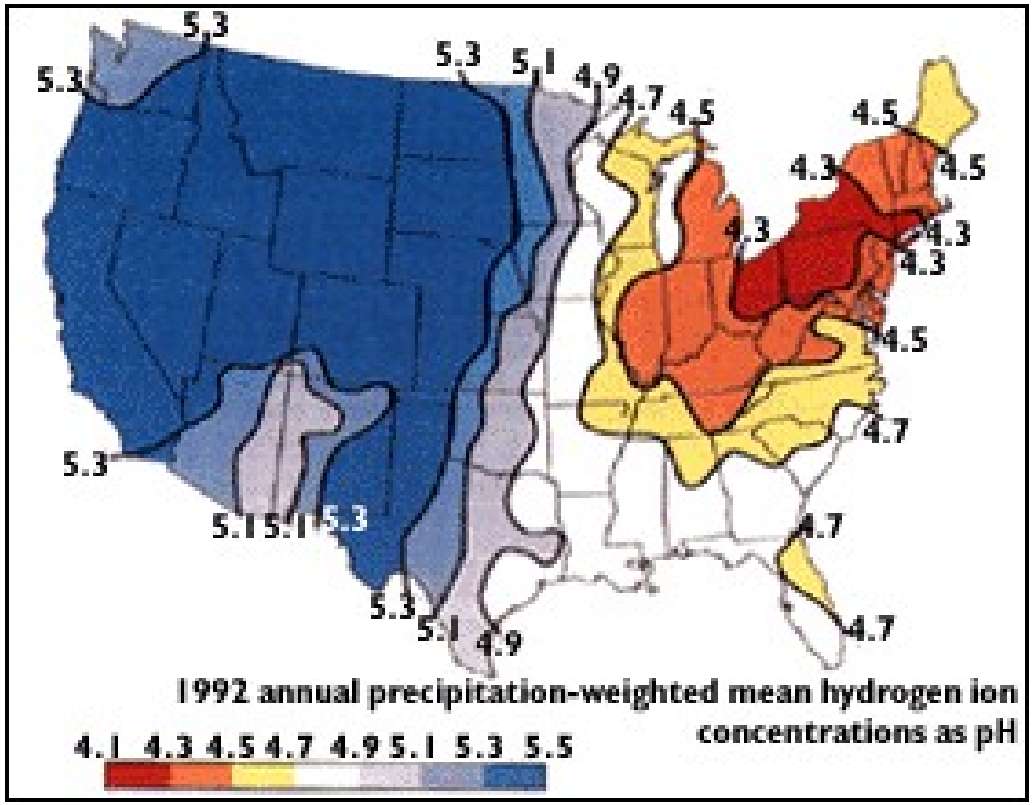
- OH is formed by a complicated set of reactions
  - photolysis of O_3 to produce atomic O, which reacts with H_2O to give OH
- OH is basically the "cleaning agent" of the atmosphere
- Example: OH reacts with NO_2
- If OH was not present, NO_2 (which is toxic) would continue to build up to a very dangerous level



Sources of NO_2 to the atmosphere

- OH reacts with NO_2 to form HNO_3
- $OH + NO_2 \rightarrow HNO_3$
- HNO_3 is then rapidly scavenged by rain droplets
- Leads to acid rain (which is another problem)



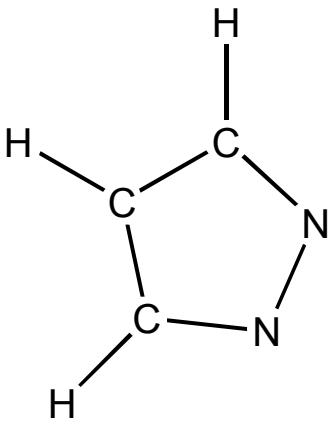


A pH distribution map shows areas in the continental United States of greatest acidity in the rain. A large part of the acidity is due to HNO3 from chemistry above.

Exercise

- Draw all Resonance structures for the **phosphate** ion and the **pyrazolate** ion which skeletal structure is shown below:

$[C_3H_3N_2]^-$



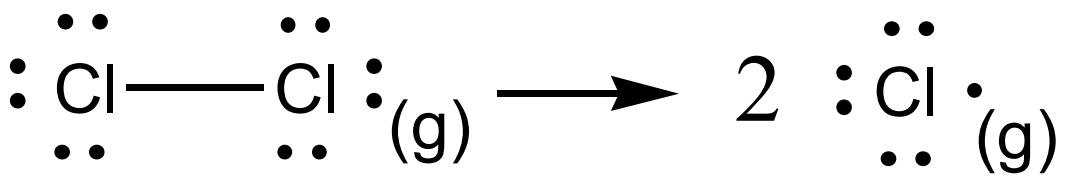
Energetics of Covalent Bonding

Define bond dissociation energies: **d**$_{o}$

Bond Strength

Bond Dissociation Energy (D_o)

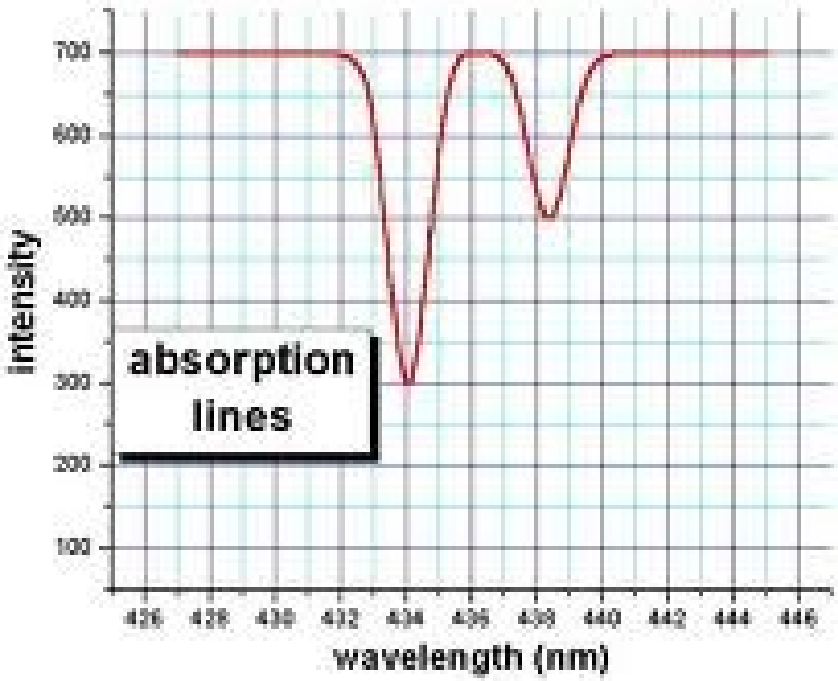
It costs energy to break bonds!



$D_o = +243$ kJ/mol

- **The higher the bond order, the shorter the bond length, the higher the bond energy.**
- Trend in D_o: Cl-Cl > Br-Br > I-I

(correlates to size)

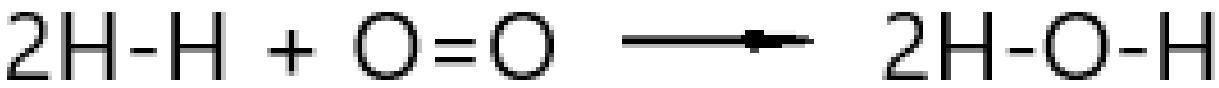


SCISAT-1 - Canadian Space Agencywww.asc-csa.gc.ca
The graph above shows an absorption spectrum.

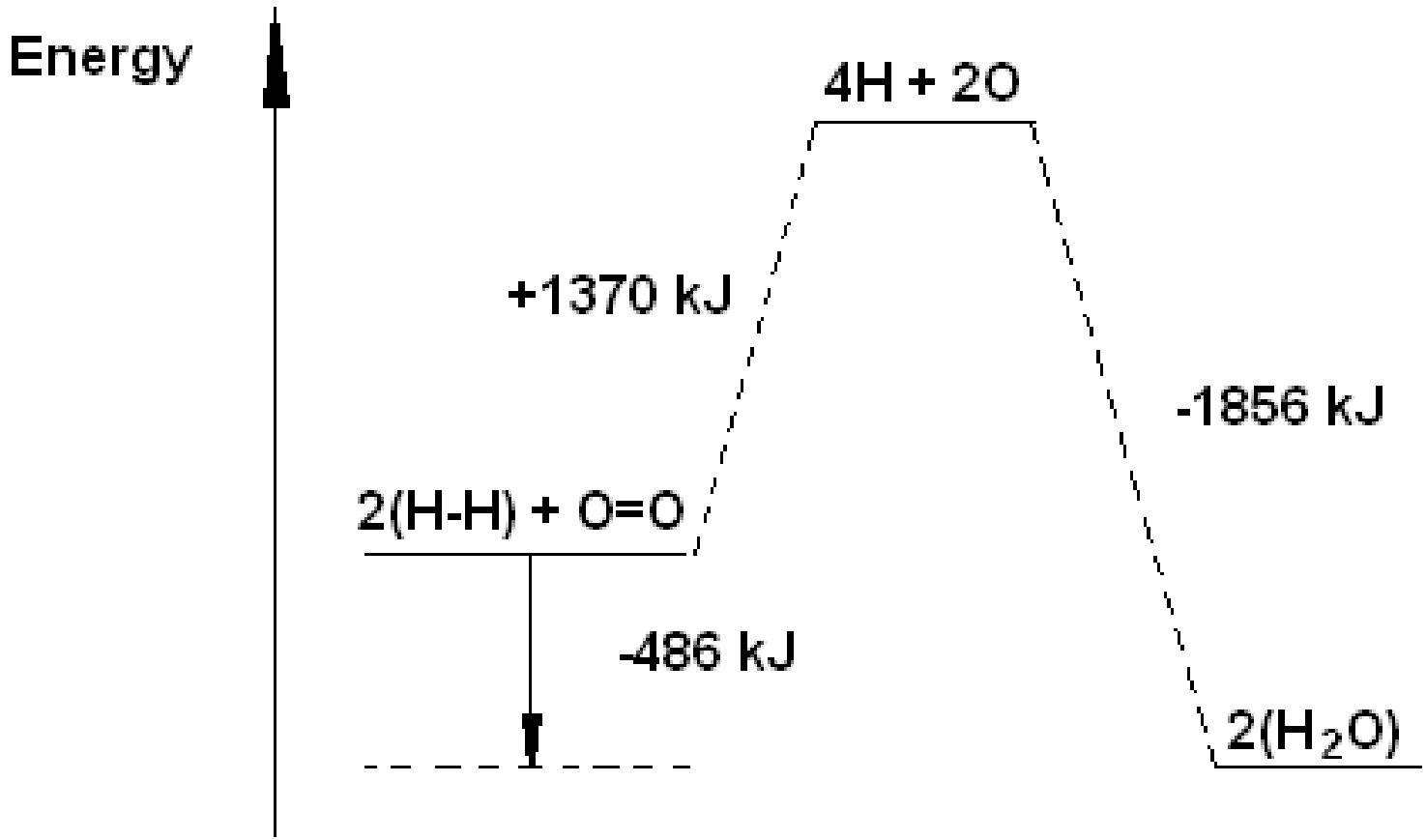
Class Exercise

- Estimate the energy change that occurs when two mol of hydrogen combine with one mol of oxygen to form water in the gas phase.
- $\mathbf{2H_2(g) + O_2(g) \rightarrow 2H_2O(g)}$

Strategy:



bond	bond energy (kJ/mol)	bond	bond energy D(o)	bond	D(o)
H-H	436	C-C	347	N-N	163
H-C	414	C=C	611	N=N	418
H-N	389	CΞC	837	NΞN	946
H-O	464	C-N	305	N-O	222
H-S	368	C=N	615	N=O	590
H-F	565	CΞN	891	O-O	142
H-Cl	431	C-O	360	O=O	498
HBr	364	C=O	736	F-F	159
H-I	297	C-Cl	339	Cl-Cl	243
Br-Br	193	I-I	151		



$$2H_2(g) + O_2(g) \rightarrow 486\ kJ + 2H_2O(g)$$

Class exercise and demo

Estimate the energy change that occurs when 20 g isopropanol CH3-CH(OH)-CH3 combust quantitatively to carbon dioxide and water.